

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented a **Java Spring Boot** service to manage **geotechnical data ingestion**, processing approximately **1,500 data points per hour** from various sensor inputs.
- Developed and maintained **Hibernate** ORM mappings for the **soil mechanics data model**, ensuring efficient persistence and retrieval operations against a **PostgreSQL** database.
- Authored **JUnit** and **Mockito** test suites for core Java components, achieving over **90% code coverage** and reducing regressions in production deployments.
- Contributed to technical proposal development for new client projects, including drafting sections on **geotechnical analysis methodologies** and required software tooling.
- Participated in **agile ceremonies** and sprint planning sessions, providing technical estimations for task completion and collaborating on backlog grooming for the **tailings management team**.
- Containerized **Java applications** using **Docker**, facilitating consistent deployment across development, testing, and production environments on **AWS EC2** instances.
- Provided **production support** for enterprise applications, responding to P1/P2 incidents and performing root cause analysis using **Splunk** and **Dynatrace** monitoring tools.

SKILLS

Engineering Domain – soil mechanics, geotechnical engineering, tailings management facilities, waste rock dump design, soil improvement methods, geosynthetic materials engineering, civil engineering

Programming Languages – Java, Python, SQL, R

Frameworks & Libraries – Spring Boot, Hibernate, JUnit, Mockito, pandas, NumPy, scikit-learn, FastAPI

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Tools & Platforms – Power BI, Tableau, Splunk, Dynatrace, Jira, Confluence

PROJECTS

Geotechnical Data Analysis Platform (HCL Technologies)

- Developed a **Python**-based data pipeline to ingest and process **geotechnical survey data**, handling approximately **500MB of raw data** daily.
- Engineered **feature extraction** logic using **pandas** and **NumPy** to derive key soil properties relevant to slope stability analysis.
- Created interactive **Power BI dashboards** to visualize soil stratification, shear strength parameters, and potential risk factors for engineering review.

Environmental Monitoring System (MSc Project)

- Designed and implemented a **Java Spring Boot** application to ingest real-time environmental sensor data, simulating inputs from water impoundments.
- Built RESTful APIs for data retrieval and analysis, integrating with a **PostgreSQL** backend to store time-series readings.
- Developed automated alerts for abnormal readings, such as rapid water level changes, reducing manual monitoring tasks by approximately **40%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)